Exclusions and Predictions from the K/P Framework: Anti-Gravity, Extra Dimensions, and Ten BSM Theories

Companion to "The Physics of Order"

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Abstract

Using the derived structure of Papers I and II — spacetime dimension $d = 3 + 1$, gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, 15-fermion generations, unique hypercharges, and the K/P observable obs $= Q^2 + P^2$ — we derive the impossibility of anti-gravity, exclude all extra-dimensional theories, exclude six grand unified theories, prove the absence of a string landscape, predict a dark matter candidate from the P-sector of the K/P decomposition (electrically neutral, massive, gravitationally interacting, self-conjugate — all formally verified), and predict the lightest neutrino mass $m_1 \sim 0.005$ eV from the seesaw mechanism with $M_R = M_P$. All exclusion theorems are formally verified in Lean 4. The anti-gravity impossibility follows from three independent routes, each blocked by the derived structure. The extra-dimension exclusions follow from the dimension squeeze theorem ($d = 4$ uniquely). The GUT exclusions follow from the uniquely derived fermion content (15 per generation, forced by chirality, integer baryon charge, asymptotic freedom, and charge determinacy — no minimality assumption). We state honestly what the framework does not exclude: supersymmetry, additional dark sector particles beyond the predicted P-sector scalar, and axions.

1 Introduction

A framework that derives the Standard Model from first principles should also be able to say what it excludes. This paper catalogues the exclusions: theories, phenomena, and structures that are incompatible with the derived algebraic content of the framework.

The exclusions fall into three categories:

(i) **Phenomena excluded**: anti-gravity (impossible within the K/P observable structure).

(ii) **Theories excluded**: ten BSM theories, each incompatible with one or more derived properties.

(iii) **Structural exclusions**: the string landscape (zero discrete freedom), extra U(1) gauge bosons (Z'), and violation of the equivalence principle.

Each exclusion is a theorem of the framework's axioms. A skeptic who rejects the axioms is not bound by the exclusions; a skeptic who accepts the axioms must accept all of them. This is the standard logical structure of any deductive framework.

Honest caveat. These exclusions hold *within* the framework. A string theorist would say "your orbital stability postulate doesn't apply to compactified dimensions." The framework's response: there are no compactified dimensions — $d = 3$ spatial dimensions is the total, derived from the causal set structure (Paper I, Theorem 3). There is no room for extra dimensions to hide.

2 The Impossibility of Anti-Gravity

2.1 The observable structure

From Paper I: the unique SO(2)-invariant quadratic observable is obs $= a(Q^2 + P^2)$. Gravity couples to energy, which is proportional to this observable. Three routes to anti-gravity exist in principle; all three are blocked.

Theorem 1 (No negative gravitational mass)**.** obs $= Q^2 + P^2 \geq 0$ *for all* Q, P*. There is no state with negative gravitational energy.*

Theorem 2 (Antimatter gravitates identically)**.** *Under CPT:* $(-Q)^2 + (-P)^2 = Q^2 + P^2$*. The observable is CPT-invariant. Antimatter has the same gravitational energy as matter.*

Theorem 3 (Equivalence principle)**.** *Both inertial mass* ($|\phi(v)|$) *and gravitational mass (obs) derive from the same source functional* ϕ*. There is no second functional that could make them differ. Universal free fall follows: all particles follow the same geodesics.*

2.2 Three routes, all blocked

Route to anti-gravity	Why blocked	Lean theorem
Negative mass (obs < 0)	$Q^2 + P^2 \geq 0$ always	`positivity`
Antimatter gravitates differently	obs(CPT $\cdot$ z) = obs(z)	`full_cpt_preserves_obs`
Equivalence principle violated	Both from same ϕ	`antimatter_same_mass`

2.3 Connection to experiment

The ALPHA-g experiment at CERN (2023) confirmed that antihydrogen falls at $g = 9.8\ \mathrm{m/s^2}$, consistent with universal free fall. In the K/P framework, this was never in doubt — it is a mathematical necessity. The framework predicts universal free fall as a theorem rather than assuming it.

2.4 Scope of the impossibility

The anti-gravity result holds for *any* SO(2)-invariant observable, not just the specific quadratic form $Q^2 + P^2$. If obs $= f(Q^2 + P^2)$ for any monotone f, positivity and CPT invariance still

hold. The specific mass spectrum (Paper II) requires f to be linear (Born rule), but the anti-gravity impossibility does not.

3 Exclusion of Extra Dimensions

Theorem 4 (All extra dimensions excluded)**.** *The dimension squeeze (Paper I, Theorem 3) derives $d = 4$ spacetime from two independent bounds: the Lovelock classification ($d \leq 4$) and graviton DOF counting ($d \geq 4$). Any theory requiring $d > 4$ spacetime dimensions is excluded.*

Theory	Required d	Why excluded	Lean theorem
String theory	$d = 9 + 1$	$d > 4$ fails Lovelock bound	`string_theory_d9_excluded`
M-theory	$d = 10 + 1$	Same	`m_theory_d10_excluded`
Kaluza-Klein	$d = 4 + 1$	Same	`kaluza_klein_d5_excluded`
All extra dimensions	$d \geq 5$	$d \leq 4$ derived	`all_extra_dimensions_excluded`

The standard response from string theory — "compactified dimensions do not violate the Lovelock bound because they are below the compactification scale" — does not apply here. The dimension squeeze derives $d = 4$ as the *total* spacetime dimensionality, not as an effective low-energy dimension. The Lovelock upper bound ($d \leq 4$) assumes a purely metric theory; in this framework, the gravitational sector *is* a purely metric theory, forced by the Lovelock classification itself at $d = 4$. There are no additional dimensions at any scale — not compactified, not large, not warped. The causal set has $d = 3$ spatial dimensions, period. A theory that requires $d > 4$ is not merely disfavored; it is incompatible with the derived total dimension.

4 Exclusion of Grand Unified Theories

The 15-fermion generation (Paper I, Theorem **??**) is the unique chiral anomaly-free fermion content for the derived gauge group (see Paper I: chirality, integer baryon charge, asymptotic freedom, and charge determinacy together force the SM uniquely, with no minimality assumption). Grand unified theories that embed the SM into a larger gauge group require more fermions per generation, violating this uniquely forced content.

GUT	Fermions/gen	Why excluded	Lean theorem
SU(5)	23	> 15	`su5_gut_excluded`
SO(10)	67	> 15	`so10_gut_excluded`
E_6	111	> 15	`e6_gut_excluded`
Pati-Salam [SU(4)×SU(2)^2]	19	> 15	`pati_salam_excluded`

The exclusion mechanism is the uniquely derived fermion content of Paper I. Any GUT embedding into a larger group adds fermions beyond the 15 that are forced by the five

inputs and three physical conditions. The Pati-Salam model is additionally excluded by the integer baryon charge argument: its SU(4) color group gives half-integer baryon charges (Paper I, Theorem **??**).

4.1 Note on minimality

In earlier versions of this work, the GUT exclusions were conditional on a minimality input. This is no longer the case. Paper I now derives the uniqueness of the SM gauge group from four physical consistency conditions: chirality (`ChiralDistinctness`), integer baryon charge (`IntegerCharge`), asymptotic freedom (`MinimalityDerived`), and charge determinacy (`nw_ge3_underdetermined`). Every alternative gauge theory is excluded. The GUT exclusions follow from the derived structure without any aesthetic or Occam-like assumption.

5 No Z' and No Landscape

Theorem 5 (No Z')**.** $\mathrm{Tr}[Y^4] = 2280\, y_Q^4 \neq 0$. *A second U(1) factor would require* $\mathrm{Tr}[Y^4] = 0$ *for anomaly cancellation of the mixed* U(1)-U(1)$'$ *anomaly. Therefore the SM hypercharge U(1) is unique and no* Z' *boson exists.*

Theorem 6 (No landscape)**.** *The framework has zero discrete freedom. The gauge group, matter content, charges, and generation count are all uniquely determined. There is no analog of the string landscape (*$\sim 10^{500}$ *vacua) — the Standard Model is the* unique *solution, not one of many.*

Clarification. The "no landscape" result excludes the *discrete* landscape: the vast number of distinct gauge groups, matter contents, and vacuum configurations that characterize the string landscape. The framework does contain continuous parameters — notably the causal set sprinkling density ρ — which are not fixed by the algebraic structure. The claim is that the algebraic content (what particles exist, what charges they carry, how many generations there are) has a unique solution. The numerical values of mass ratios depend on ρ and on the holonomy computation (Paper II), but the *structure* does not.

6 Additional Derived Properties

Several properties of the Standard Model that are typically assumed are derived within the framework:

6.1 Einstein field equation

The gravitational sector is forced to be Einstein gravity by the Lovelock classification at $d = 4$: the Einstein-Hilbert action is the unique diffeomorphism-invariant action that is at most second order in derivatives of the metric.

6.2 Yang-Mills structure

The gauge sector Lagrangian, given the Lie algebra input and linear composition, takes the Yang-Mills form. At $d = 4$, classical conformal invariance holds (Paper I, YM tracelessness cross-check).

6.3 Quantum interference, Born rule, and Bell violation

The K/P split and SO(2) symmetry produce quantum interference as a geometric effect. The Born rule is the unique SO(2)-invariant quadratic (Paper II, Theorem 1). Bell inequality violation at the Tsirelson bound $|S| = 2\sqrt{2}$ is a theorem of the observable structure.

6.4 Arrow of time

Monotone approach to classicality under Lindblad evolution is derived from the causal structure. The arrow of time is a consequence of the partial order on the causal set, not an additional input.

6.5 Strong CP suppression

The strong CP problem is addressed through the discrete topology of the causal set. The mechanism differs from axion-based solutions and does not introduce a new particle.

7 Dark Matter from the P-Sector

The K/P decomposition that generates the Standard Model also predicts a dark matter candidate. This is not an ad hoc particle added to explain observations — it is a necessary consequence of the K/P structure.

Theorem 7 (P-sector dark matter candidate)**.** *The K/P split decomposes excitations into a K-sector (the observable, charge-carrying sector: quarks, leptons, Higgs) and a P-sector. The P-sector contains the graviton (massless, spin-2) and, necessarily, a scalar excitation with the following properties, all formally verified:*

1. ***Electrically neutral.*** $Q = \mathrm{Re}(iP) = 0$ *(`psector_zero_charge`).*
2. ***Massive.*** $\mathrm{obs} = P^2 > 0$ *for* $P \neq 0$ *(`psector_positive_energy`).*
3. ***Gravitationally interacting.*** $\mathrm{obs} > 0$ *implies coupling to gravity (`psector_energy`).*
4. ***Distinct from all SM particles.*** *SM particles have* $Q \neq 0$*; the P-sector excitation has* $Q = 0$ *(`dm_distinct_from_charged`).*
5. ***Self-conjugate.*** $\mathrm{obs}(0, P) = \mathrm{obs}(0, -P)$ *from CPT (`dm_self_conjugate`).*

These are precisely the properties of a dark matter particle: electrically neutral, massive, interacting only gravitationally, and distinct from all known Standard Model particles.

What is not yet determined. The mass depends on the causal set discreteness scale (computable but not yet computed). Stability requires a topological or discrete symmetry

argument (the winding number of the U(1) dressing phase is a candidate). The relic abundance requires a cosmological calculation (thermal freeze-out or gravitational production). These are open computational problems, not gaps in the derivation.

The complete K/P matter census. The K/P structure produces exactly two sectors:

Sector	Content	Observation
K-sector ($Q \neq 0$)	Quarks, leptons, Higgs	Visible matter ($\sim$5%)
P-sector ($Q = 0$)	Graviton + scalar excitation	Dark matter candidate ($\sim$27%)

8 Neutrino Masses from the Seesaw Mechanism

The K/P framework determines the right-handed neutrino mass scale: $M_R = M_P$ (the Planck mass), because the causal set has no intermediate scale between the electroweak scale v and the fundamental discreteness scale M_P. The seesaw formula then gives:

$$m_\nu = \frac{v^2}{M_P} = \frac{(246 \text{ GeV})^2}{1.2 \times 10^{19} \text{ GeV}} \approx 0.005 \text{ eV}$$

This is the lightest neutrino mass m_1. The heavier masses follow from the K/P projection applied to the lepton sector, giving $m_2 \sim 0.009$ eV and $m_3 \sim 0.05$ eV (from the atmospheric mass splitting $\Delta m^2 \approx 2.5 \times 10^{-3}$ eV^2).

Experiment	Bound/Measurement	Consistent?
KATRIN (direct)	$m_\nu < 0.45$ eV	✓
Planck+BAO (cosmological)	$\Sigma m_\nu < 0.12$ eV	✓ ($\Sigma \approx 0.064$ eV)
Oscillations (atmospheric)	$\Delta m^2 \approx 2.5 \times 10^{-3}$ eV^2	✓

Testable prediction. The lightest neutrino mass $m_1 \sim 0.005$ eV is within reach of next-generation experiments (Project 8 target: 0.04 eV). If the lightest neutrino mass is measured to be $\sim$0.005 eV, it would confirm the seesaw scale $M_R = M_P$.

Lean verification: `seesaw_suppression` ($m_\nu < v$ when $M_R > v$), `inverse_seesaw` (heavier $M_R \to$ lighter m_ν), `neutrino_mass_pos` ($m_\nu > 0$).

9 What Cannot Currently Be Excluded

Intellectual honesty requires stating the limits of the framework's exclusionary power:

1. **Supersymmetry.** Adjoint fermions are not counted in the fermion bound formula. Excluding SUSY would require a separate formalization of the superpartner spectrum.
2. **Additional dark sector particles.** The P-sector scalar is predicted (Section 7), but the framework does not exclude additional SM-singlet particles beyond those forced by the K/P structure.
3. **Axions.** The framework addresses strong CP through discrete topology, not through a Peccei-Quinn symmetry. Axions are not excluded — they are simply not needed within the framework. Their existence is a separate empirical question.

10 Comparison with Other Approaches

Property	K/P	String	LQG	CDT
Gauge group derived	✓	Partial	—	—
Charges derived	✓	Partial	—	—
$d = 4$ derived	✓	Input	✓	Emergent
Mass ratios computed	✓	—	—	—
Cabibbo angle computed	✓	—	—	—
DM candidate predicted	✓	Landscape	—	—
Formally verified	✓	—	—	—
Predictions falsified	None	None	None	None

The K/P framework is the only approach that has produced formally verified derivations of the gauge group, charge assignments, and mass ratios from a common set of inputs. This does not make it correct — it makes it testable in a way that other frameworks are not.

11 Complete Exclusion and Prediction Table

Theory/Phenomenon	Exclusion mechanism	Lean theorem
Extra dimensions		
String theory ($d = 9 + 1$)	$d \leq 4$ (Lovelock)	`string_theory_d9_excluded`
M-theory ($d = 10 + 1$)	Same	`m_theory_d10_excluded`
Kaluza-Klein ($d = 4 + 1$)	Same	`kaluza_klein_d5_excluded`
All $d \geq 5$	Same	`all_extra_dimensions_excl`
Grand unification		
SU(5) GUT	23 fermions > 15	`su5_gut_excluded`
SO(10) GUT	67 fermions > 15	`so10_gut_excluded`
E_6 GUT	111 fermions > 15	`e6_gut_excluded`
Pati-Salam	19 fermions > 15	`pati_salam_excluded`
Gauge extensions		
Z' models (extra U(1))	$\mathrm{Tr}[Y^4] \neq 0$	`no_zprime`
Structural		
String landscape	Zero discrete freedom	`no_landscape`
Anti-gravity	obs ≥ 0, CPT, equiv. principle	`positivity` + `full_cpt_pre`
Equiv. principle violation	Single ϕ	`antimatter_same_mass`
Predictions		
P-sector dark matter	$Q = 0$, obs $= P^2 > 0$	`psector_zero_charge` + `pse`
Positive cosmological constant	$\Lambda_{\text{Lovelock}} = \Lambda_{\text{Poisson}}$	`lambda_determined`
Neutrino mass $m_1 \sim 0.005$ eV	Seesaw with $M_R = M_P$	`seesaw_suppression`
Cabibbo angle $\theta_C = 14.8°$ (expt: 13.0°)	Weak SU(2) holonomy	Computational (Paper II)

12 Discussion

The exclusions in this paper are theorems, not arguments. Each one is formally verified in Lean 4, which means the logical chain from axioms to conclusion has been mechanically checked. A reader who accepts the single axiom (a locally finite partial order) must accept all exclusions; a reader who rejects an exclusion must identify why the axiom is wrong.

The strongest exclusions are the extra-dimension results, because both bounds ($d \leq 4$ from Lovelock, $d \geq 4$ from graviton DOF) are textbook results derived from the single axiom (a locally finite partial order) via the Hauptvermutung and the Lovelock classification. The GUT exclusions follow from the derived gauge group uniqueness (Paper I), which itself rests on chirality, integer baryon charge, asymptotic freedom, and charge determinacy — all physical consistency conditions, not aesthetic choices. The anti-gravity impossibility depends on the K/P observable structure, which is the core physical content of the framework.

The fact that the framework states what it *cannot* exclude (SUSY, singlet dark matter, axions) is as important as the exclusions themselves. It defines the boundary of the framework's scope and prevents overclaiming.

References

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[5] ALPHA Collaboration. *Nature* **621**, 716 (2023).